Notes: Handley and Limão (AEJ: Economic Policy, 2015)

Trade and Investment under Policy Uncertainty: Theory and Firm Evidence

Contents

1 Big picture

- **Question.** Does uncertainty about future trade policy reduce firms' export-entry investment, and can a credible trade agreement raise trade even when current tariffs are already low?
- **Core mechanism.** Exporting requires a sunk investment. When a preferential tariff may be reversed in the future, firms value the option of waiting and fewer firms enter export markets today.
- **Empirical setting.** The paper studies Portugal's 1986 accession to the European Community (EC), focusing on Portuguese exports to the original EC-10 countries and Spain.
- **Key institutional fact.** For many industrial goods, Portugal already had low or zero applied tariffs before accession. The accession mattered because it made preferential access credible and reduced the probability of losing those preferences.
- **Main empirical object.** Trade-policy uncertainty is measured as the proportional profit loss a Portuguese exporter would suffer if preferences were reversed and the importing country returned to its nonpreferential tariff.
- **Main result.** Accession increased firm entry and export values. A large fraction of the predicted growth comes from removing policy uncertainty rather than from reducing applied tariffs.
- **Economic takeaway.** Trade agreements can matter not only by lowering tariffs today, but also by making future market access predictable enough for firms to sink entry and market-building costs.

2 Incremental contribution of the paper

- **A tractable model of trade-policy uncertainty.** The paper embeds policy uncertainty into a dynamic heterogeneous-firm export-entry model with sunk entry costs.
- **A concrete uncertainty measure.** Instead of using broad aggregate uncertainty indexes, it constructs product- and destination-level uncertainty from observed preferential and nonpreferential tariffs.
- **Credibility as a treatment.** The paper treats Portugal's EC accession as a shock that makes previously uncertain preferences credible.
- **Firm-level evidence.** The analysis links changes in trade policy and uncertainty to changes in the number of exporting firms and firm-varieties by industry and destination.
- **Quantification, not only signs.** The estimates are mapped back to structural objects such as the probability of policy reversal and the share of export growth attributable to tariff cuts, credibility, and complementarity.
- **New interpretation of PTAs.** The paper explains why preferential trade agreements may increase trade substantially even when measured applied tariffs change little.

3 Data and measurement

3.1 Trade and firm data

- The empirical analysis uses Portuguese firm-level export data from the Portuguese census and trade statistics.
- The key short-window analysis focuses on changes from 1985 to 1987, around the implementation

of Portugal's EC accession in 1986.

- The dependent variable in the baseline micro regressions is the log change in the number of firm-varieties exported to destination i in two-digit industry V.
- The main destinations are Spain and the EC-10. Spain joined the EC in 1986 and liberalized tariffs against Portugal, while the EC-10 had already granted broad industrial preferences to Portugal before accession.

3.2 Policy variables

The paper distinguishes three tariff concepts:

- τ_{0iV}: the preferential tariff faced by Portugal before accession.
- τ_{1iV}: the tariff faced by Portugal after accession.
- τ_{hiV}: the high, worst-case, nonpreferential tariff that Portugal could have faced if preferences were lost.

The uncertainty measure is based on the proportional profit loss from a preference reversal:

$$\omega(\tau_{tiV}) - 1 = -p_{hi}\left[1 - \left(\frac{\tau_{tiV}}{\tau_{hiV}}\right)^{\sigma}\right]. \tag{1}$$

Interpretation. If preferential access is secure, the probability of reverting to the high tariff is zero and the uncertainty term has no effect on entry. If the reversal probability is positive, the potential profit loss discourages entry.

3.3 Summary facts

- Between 1985 and 1992, Portugal's trade share with Spain and the EC-10 increased from roughly 52 percent to 72 percent.
- From 1985 to 1987, the number of Portuguese firm-varieties exported grew by about 101 log points to Spain and 26 log points to the EC-10.
- Portugal faced an initial uncertainty measure of roughly 15–16 percent profit loss if preferences were reversed.
- Spain involved both an applied tariff reduction and an uncertainty reduction. The EC-10 case is especially informative because many applied tariffs were already zero, so the main channel was credibility.

4 Model and empirical specification

4.1 Operating profits

With CES demand and monopolistic competition, the operating profit of a Portuguese firm with marginal cost c_v exporting to destination i in industry V is

$$\pi(c_v, \tau_{iV}) = \tau_{iV}^{-\sigma} A_{iV} c_v^{1-\sigma}, \tag{2}$$

where A_{iV} summarizes foreign demand and σ is the elasticity of substitution.

Interpretation. Higher tariffs lower operating profits. More productive firms, with lower c_v, have higher profits and are more likely to export.

4.2 Deterministic export-entry cutoff

If policy is deterministic, a firm enters if the present value of export profits exceeds sunk cost K_{iV}. The deterministic marginal entrant satisfies

$$c_{iV}^{D} = \left[\frac{\tau_{iV}^{-\sigma} A_{iV}}{(1-\beta) K_{iV}} \right]^{1/(\sigma-1)} . \tag{3}$$

Interpretation. A tariff reduction raises c_{iV}^{D}, so less productive firms can profitably enter.

4.3 Policy uncertainty and the entry cutoff

With policy uncertainty, the entry cutoff equals the deterministic cutoff multiplied by an uncertainty factor:

$$c_{iV}^{U} = U_t c_{iV}^{D}, \tag{4}$$

where

$$U_t = \left[\frac{1-\beta(1-\gamma\omega(\tau_t))}{1-\beta(1-\gamma)} \right]^{1/(\sigma-1)} \leq 1. \tag{5}$$

Here γ is the probability of a policy shock and $\omega(\tau_t)$ captures the expected profit effect of a policy shock.

Interpretation. If there is no policy uncertainty, $U_t = 1$. If a preference reversal is possible, $U_t < 1$ and fewer firms enter. Current tariff cuts and credibility are complements because tariff reductions have larger effects when firms believe the low tariff will persist.

4.4 Baseline estimating equation

Assuming a Pareto productivity distribution and differencing before and after accession gives the baseline empirical equation:

$$\Delta_t \ln n_{tiV} = b_{\gamma 1} \widehat{\omega}_{1iV} - b_{\gamma 0} \widehat{\omega}_{0iV} + b_\tau \Delta_t \ln \tau_{tiV} + a_i + a_V + \Delta_t u_{iV}. \tag{6}$$

- $\Delta_t \ln n_{tiV}$ is growth in exporting firm-varieties.
- $\widehat{\omega}_{0iV}$ is pre-accession profit loss if preferences are reversed.
- $\widehat{\omega}_{1iV}$ is post-accession profit loss if preferences are reversed.
- $\Delta_t \ln \tau_{tiV}$ is the applied tariff change.
- a_i and a_V are destination and industry effects.

Interpretation. The pre-accession uncertainty coefficient tests whether entry increased more in industries where the initial potential loss from preference reversal was larger. The post-accession uncertainty coefficient tests whether accession eliminated the perceived reversal risk.

5 Identification and empirical design

5.1 Identification logic

- The paper compares export-entry growth across destinations and industries with different tariff changes and different initial uncertainty exposure.

- The short window, 1985–1987, reduces contamination from slower macro changes, infrastructure investment, or later monetary integration.
- Destination fixed effects absorb aggregate destination shocks, such as Spain-specific income growth or exchange-rate effects.
- Industry fixed effects absorb common Portuguese supply shocks and industry-specific trends.
- The key variation comes from the fact that EC-10 and Spain had different preferential and nonpreferential tariff schedules across industries.

5.2 What would threaten the design?

- **Other policy changes.** If nontariff barriers or specific tariffs changed at the same time and were correlated with the uncertainty measure, the estimated uncertainty effect could be biased.
- **Foreign demand shocks.** If Spain or EC-10 demand changed differentially in high-uncertainty industries, that could mimic an uncertainty effect.
- **Elasticity assumptions.** The uncertainty measure depends on the elasticity of substitution σ.
- **Unobserved beliefs.** The reversal probability is not directly observed and must be inferred from entry responses.

5.3 Robustness checks

The paper addresses these concerns by adding controls for changes in nontariff measures, specific tariffs, applied tariff standard deviations, and a price-index proxy; by testing alternative values of σ; by using both firm and firm-variety entry; and by using alternative pre-accession periods.

6 Tables and figures

6.1 Figure 1: Portuguese trade shares with EC-10 and Spain

Read:

- The figure plots Portugal's trade share with the EC-10 and Spain from 1951 to 1992.
- The share is roughly flat near 50 percent after the earlier preferential agreements in the 1970s and early 1980s.
- The share rises sharply after the 1985 accession agreement and reaches about 72 percent by 1992.

Economic message: The macro pattern is consistent with a credibility channel. Earlier tariff preferences did not trigger a large trade-share reallocation, but the formal accession did.

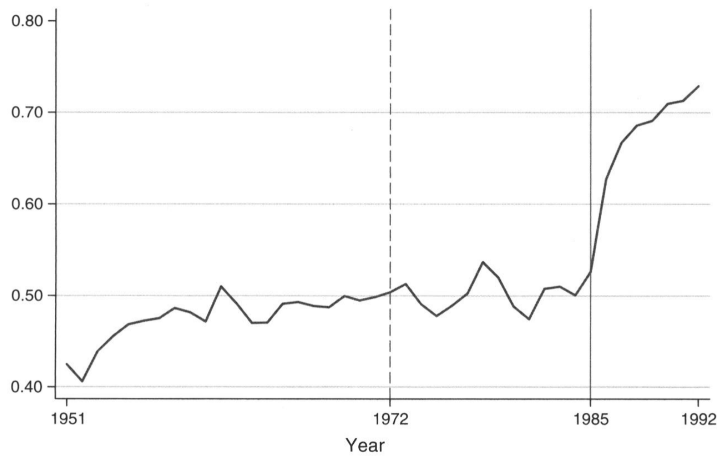

FIGURE 1. PORTUGUESE TRADE SHARES WITH EC-10 AND SPAIN

: 1972: EC industrial preferences agreed. 1985: Accession signed.

Figure 1: Portuguese trade shares with EC-10 and Spain

6.2 Table 1: Portuguese export growth margins and Table 2: Summary statistics for baseline regressions

Read:

- **Table 1** shows aggregate export growth margins from 1981 to 1990.
- After accession, Portuguese exports to the EC-10 rise by 0.232 log points, and the number of firms exporting to the EC-10 rises by 0.451 log points.
- The Spain effect is much larger: exports rise by 1.146 log points and the number of firms rises by 1.159 log points.
- Average exports per firm fall for the EC-10, consistent with many smaller firms entering.
- **Table 2** summarizes the micro sample and policy variables. Firm entry and export growth are much larger for Spain, where applied tariffs also fall sharply.
- Initial uncertainty is economically large for both Spain and the EC-10: losing preferences would cut profits by roughly 15–16 percent on average.

Economic message: The accession boom operates strongly through the extensive margin. Spain combines tariff reductions with credibility, while EC-10 is especially useful for identifying credibility because many applied tariffs were already low.

TABLE 1—PORTUGUESE EXPORT GROWTH MARGINS 1981–1990

Dependent variable (ln):	Exports (1)	Number of firms (2)	Exports/firm (3)
EC10 × Post_86	0.232***	0.451***	−0.219***
	[0.0829]	[0.0411]	[0.0710]
Spain × Post_86	1.146***	1.159***	−0.0129
	[0.199]	[0.132]	[0.113]
Real importer GDP (ln)	1.045***	0.598***	0.447*
	[0.306]	[0.137]	[0.258]
Importer price index (ln)	0.167**	0.0185	0.148**
	[0.0776]	[0.0374]	[0.0655]
Exchange rate (ln)	0.211***	−0.00118	0.212***
	[0.0763]	[0.0365]	[0.0653]
Observations	1,305	1,305	1,305
Adjusted R^2	0.92	0.97	0.75

Notes: Includes dummies for country and year. Robust standard errors are in brackets. The sample is aggregate values to each country of destination. Column 3 is the logarithm of total exports over number of firms. For variable definitions and sources see Appendix C. See online Appendix, Table D4 for summary statistics.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

Table 1: Portuguese export growth margins

TABLE 2—SUMMARY STATISTICS FOR FIRM-LEVEL BASELINE REGRESSIONS

	Total	Spain	EC-10
Change in number of firms[a]	33.0	91.1	24.7
	(55.1)	(62.6)	(48.7)
Change in number of varieties[a]	35.7	101	26.4
	(60.7)	(69.1)	(53.3)
Change in exports[a]	55.3	135	43.9
	(157)	(150)	(155)
Pre-tariff[b] (faced by Portugal)	3.13	7.89	2.45
	(5.66)	(5.10)	(5.40)
Pre-tariff[b] (faced by GATT countries)	9.14	14.07	8.44
	(5.29)	(7.75)	(4.43)
Post-tariff[b] (faced by Portugal)	1.74	1.33	1.79
	(3.91)	(3.51)	(3.96)
Tariff change[b] (faced by Portugal)	−1.39	−6.56	−0.66
	(2.90)	(4.78)	(1.44)
Applied tariff standard deviation change[c]	−0.64	−2.86	−0.33
	(1.25)	(1.86)	(0.69)
Price index proxy change[c]	−0.19	−1.52	0.00
	(0.88)	(2.06)	(0.00)
NTM share change[c]	−2.32	−18.66	0.00
	(10.9)	(25.4)	(0.00)
Specific tariff share change[c]	−0.37	−3.01	0.00
	(2.80)	(7.45)	(0.00)
Proportion of profits lost if preference reversed			
Initial uncertainty (at pre-tariffs)	15.5	16.0	15.4
	(10.9)	(9.52)	(11.1)
Post uncertainty (at post-tariffs)	18.8	29.4	17.3
	(10.8)	(15.0)	(9.08)
Observations	731	91	640

Note: Sample means and standard deviations in parentheses.

[a] $100 \times \Delta \ln(x)$, where $x = \{\text{firms, varieties, exports}\}$.

[b] $100 \times \ln(1 + t)$, where t is the ad valorem rate; "Pre-tariff" is evaluated in 1985 (pre-accession); "Post-tariff" is the 1987 tariff faced by Portugal; "Tariff change" is a simple difference.

[c] See Appendix C for sources and details. Profit loss term is $1 - (\tau_{0V}/\tau_{hV})^{\sigma}$ ($\sigma = 3$). Loss measures in regressions divided by $\sigma - 1$ to normalize.

Table 2: Summary statistics for baseline regressions

6.3 Table 3: Firm-variety entry growth into EC-10 and Spain and Table 4: Reversal, attenuation, and entry estimates

Read:

- **Table 3** estimates the baseline firm-variety entry equation.
- Initial uncertainty has a positive and significant coefficient across specifications, meaning entry grew more in industries where the pre-accession loss from preference reversal was larger.
- Applied tariff changes have a negative coefficient: tariff reductions increase entry, as the model predicts.
- Post-accession uncertainty is insignificant, consistent with accession eliminating the perceived probability of preference reversal.
- **Table 4** maps the estimates into structural quantities. The baseline probability of reversal is about 0.39 when $\sigma = 3$.
- The tariff elasticity attenuation factor is about 0.56, meaning uncertainty cuts the entry response to tariff reductions roughly in half.

Economic message: Table 3 is the core reduced-form/structural evidence. Table 4 shows why credibility matters quantitatively: under uncertainty, tariff cuts are much less powerful because firms fear that market access may not persist.

TABLE 3—FIRM-VARIETY ENTRY GROWTH INTO EC-10 AND SPAIN
(By industry)

	Change in (ln) number of firm varieties					
Dependent variable:	(1)	(2)	(3)	(4)	(5)	(6)
Initial uncertainty ($-b_{\gamma 0} > 0$)	4.399** [1.772]	5.626** [2.756]	4.301** [1.810]	4.431** [1.788]	4.351** [1.839]	4.752** [1.854]
Applied tariff change ($b_\tau < 0$)	−3.006** [1.260]	−4.273* [2.271]	−3.072** [1.266]	−2.919** [1.247]	−3.113** [1.291]	−3.520*** [1.260]
Post uncertainty ($b_{\gamma 1} \leq 0$)		−1.51 [3.277]				
NTM share change			−0.166 [0.256]			
Specific tariff share change				−0.579 [1.034]		
Applied tariff SD change					0.468 [4.144]	
Price index Proxy change						1.946 [2.066]
Observations	731	731	731	731	731	731
R^2	0.471	0.471	0.472	0.472	0.471	0.471
Number of parameters	101	102	102	102	102	102
AIC	1,083	1,085	1,084	1,084	1,085	1,084
BIC	1,551	1,558	1,557	1,557	1,558	1,558

Notes: Structural parameters and expected sign in parentheses below regressor names. All specifications include country and industry effects. Clustered standard errors are in brackets (industry × EC-10). The sample is Spain and EC 10 countries, 1987–1985. All regressions assume that $\sigma = 3$. AIC and BIC denote Akaike and Bayes Information Criterion. See Table 2 for summary statistics.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Table 3: Firm-variety entry growth into EC-10 and Spain

TABLE 4—REVERSAL, ATTENUATION, AND ENTRY ESTIMATES

Value of $\sigma =$	2	3	4
Probability of reversal	0.36	0.39	0.42
(standard error)	(0.16)	(0.17)	(0.18)
Tariff elasticity attenuation factor	0.56	0.56	0.55
Entry growth, uncertainty removal (mean ω)	0.08	0.08	0.08
Entry growth, uncertainty removal (min. ω)	0.24	0.24	0.24

Notes: We use the initial uncertainty estimate from Table 3 in calculations. Probability of reversal $= \frac{\hat{b}_{\gamma 0}}{\hat{b}_\tau} \frac{\sigma}{\sigma - 1} \frac{1-\beta}{\beta}$ with standard error obtained using delta method ($\beta = 0.85$). Conditional on some p_h, the attenuation and theoretical uncertainty term U can be derived using regression estimates independently of β assumptions. We take $p_h = (\text{Pr. reversal})^{0.5}$ and compute attenuation and entry growth at mean ω. The attenuation factor is the ratio of the entry elasticity to tariff changes at initial uncertainty relative to no uncertainty; see equation (24). We compute entry due to uncertainty removal as the log difference in the number of entrants at post-uncertainty versus initial uncertainty using $k \times [\ln(U_1) - \ln(U_0)]$ and assume post-uncertainty is removed, $U_1 = 1$.

Table 4: Reversal, attenuation, and entry estimates

6.4 Table 5: Probability of reversal robustnes and Table 6: Entry counterfactuals and quantificatio

Read:

- **Table 5** shows that the inferred probability of reversal is robust to different elasticities and to using firm counts instead of firm-variety counts.
- The baseline 1987–1985 sample gives reversal probabilities around 0.36–0.42 depending on σ.
- Using the average of 1983–1985 as the preperiod lowers the estimate but still implies meaningful reversal risk.
- **Table 6** decomposes predicted entry growth into uncertainty removal, applied tariff reduction, and complementarity.
- Overall predicted entry growth is 0.12 log points; uncertainty removal accounts for 65 percent, tariff reductions for 19 percent, and complementarity for 16 percent.
- For EC markets, uncertainty removal accounts for 80 percent of the predicted effect; for Spain, tariff cuts are more important but credibility and complementarity remain large.

Economic message: The estimates are not driven by one elasticity assumption or one entry definition. The decomposition shows that a noncredible tariff reduction would have produced much smaller entry effects.

TABLE 5—PROBABILITY OF REVERSAL ROBUSTNESS

Sample	1987–1985		1987–pre-mean	
Dependent variable (ln), number of:	Varieties	Firms	Varieties	Firms
Value of $\sigma = 2$	0.36	0.36	0.27	0.26
3	0.39	0.39	0.29	0.28
4	0.42	0.41	0.31	0.30

Notes: We assume $\beta = 0.85$, for any other value of β, simply divide by $0.15/0.85$ and multiply by the new $(1 - \beta)/\beta$. The 1987–1985 sample uses the growth between 1987 and 1985, which is the baseline. The 1987–pre-mean sample uses the growth between 1987 and the average of the three years before the agreement: 83, 84, 85.

Table 5: Probability of reversal robustness

TABLE 6—ENTRY COUNTERFACTUALS AND QUANTIFICATION

	Total	Spain	EC
Predicted entry effect of tariff and uncertainty reductions	0.12	0.28	0.10
Share of predicted effect due to:			
Uncertainty removal (at initial tariffs)	0.65	0.28	0.80
Tariff reduction (at initial uncertainty)	0.19	0.40	0.11
Complementarity	0.16	0.32	0.09
Share of predicted entry probability explained by policy	0.61	0.33	1.03

Notes: We use the initial uncertainty estimate from Table 3 in calculations (but results are similar if $\sigma = 2, 4$). Predicted average entry probability is the sum tariffs reductions, $b_\tau \Delta \ln \tau$, and removal of uncertainty at initial tariffs. See Appendix B for further details. Entry growth from uncertainty removal is the log difference in the number of entrants at post-uncertainty versus initial uncertainty using $k \times [\ln(U_1) - \ln(U_0)]$ (we assume post-uncertainty is removed, $U_1 = 1$). Counterfactual shares of predicted entry hold initial tariffs and uncertainty fixed, respectively. The complementarity share captures the remaining entry growth from simultaneously reducing tariffs and uncertainty. The share of predicted entry probability explained by policy is the ratio of policy predictions relative to the one observed in the data.

Table 6: Entry counterfactuals and quantification

6.5 Table 7: Credibility effect from securing preexisting reforms and Table 8: Total exports counterfactuals and quantification

Read:

- **Table 7** focuses on the credibility effect from securing preexisting reforms.
- The total credibility effect is 8.4 log points for entry and 21.2 log points for exports.
- More than half of the entry credibility effect, and about half of the export credibility effect, is due to pure risk reduction rather than simply locking in a lower mean tariff.
- **Table 8** extends the counterfactual analysis to export values.
- Predicted export growth from policy is 0.34 log points overall, and policy explains 87 percent of the observed export growth net of growth in the number of Portuguese firms.
- Uncertainty removal accounts for 59 percent of predicted export growth overall, and 75 percent for the EC-10.

Economic message: Credibility matters not just for the number of exporters, but also for export values. The export effects are especially striking because the policy variables explain most of the observed export growth in the short window.

TABLE 7—CREDIBILITY EFFECT FROM SECURING PREEXISTING REFORMS

	Total	Spain	EC
Entry growth:			
Total effect (ln points)	8.4	8.6	8.3
Share from risk reduction (at long-run mean)	0.57	0.83	0.53
Export growth:			
Total effect (ln points)	21.2	21.8	21.1
Share from risk reduction (at long-run mean)	0.50	0.73	0.47

Notes: Calculations use an initial uncertainty estimate from the entry or total export regressions where $\sigma = 3$ and the probability of reversal is estimated at $r_0 = 0.39$ and $r_0 = 0.45$, respectively. Total growth from uncertainty removal is the log difference in entry or exports at post versus initial uncertainty using k or $(k - \sigma + 1)$ times $[\ln(U_1) - \ln(U_0)]$ evaluated at the preperiod applied (τ_0) and GATT tariff (τ_h) means from Table 2 (we assume post-uncertainty is removed, $U_1 = 1$). The counterfactual shares from risk reduction evaluate $\ln(U_0)$ while holding applied tariffs at $\tau_0 = E(\tau)$ to obtain a mean preserving compression of the tariff distribution. We compute $E(\tau)$ from our estimates of α, η (results are similar if $\sigma = 2, 4$).

Table 7: Credibility effect from securing preexisting reforms

TABLE 8—TOTAL EXPORTS COUNTERFACTUALS AND QUANTIFICATION

	Total	Spain	EC
Predicted export effect of tariff and uncertainty reductions	0.34	0.85	0.26
Share of predicted total exports due to:			
Uncertainty removal (at initial tariffs)	0.59	0.23	0.75
Tariff reduction (at initial uncertainty)	0.28	0.53	0.17
Complementarity	0.13	0.24	0.08
Share of predicted total exports explained by policy	0.87	0.72	0.96

Notes: Calculations use an initial uncertainty estimate from the total export regression where $\sigma = 3$ and the probability of reversal is estimated at 0.45; see Appendix B for details (results are similar if $\sigma = 2, 4$). Average predicted total exports are the sum of the tariff reduction, $B_\tau \Delta \ln \tau$, and the effect of uncertainty removal at initial tariffs. Total export growth from uncertainty removal is the log difference in exports at post-uncertainty versus initial uncertainty using $(k - \sigma + 1) \times [\ln(U_1) - \ln(U_0)]$ (we assume post-uncertainty is removed, $U_1 = 1$). Counterfactual shares of predicted exports hold initial tariffs and uncertainty fixed, respectively. The complementarity share captures the remaining export value growth from simultaneously reducing tariffs and uncertainty. The share of total export growth explained by policy is the ratio of policy predictions relative to exports observed in the data net of the aggregate

Table 8: Total exports counterfactuals and quantification

6.6 Data appendix table

Read:

- The appendix table defines the aggregate variables, firm-entry measures, tariff measures, uncertainty measure, and auxiliary controls.
- The uncertainty variable is defined as the proportional reduction in per-period profits if the tariff faced by an exporter reverts from the preferential tariff to the nonpreferential tariff.
- The table also clarifies that the empirical entry objects are changes in the number of firms and firm-varieties by destination and two-digit industry.

Economic message: The appendix table is useful because the credibility channel depends on carefully constructed policy variables. The paper's contribution is not just theoretical; it also builds a dataset that makes policy uncertainty measurable at the relevant firm-industry-destination level.

C. *Data Appendix*

DATA APPENDIX TABLE

Variable	Description	Source
Aggregate regressions (Table 1, 1981–1990):		
Exports (ln)	Nominal value of exports in euro of all goods to country i in year t	a
Number of firms exporting (ln)	Number of uniquely identified shippers with positive exports to i in year t	a
Exports per firm (ln)	ln (Exports$_{it}$/Number of firms$_{it}$)	a
Real importer GDP (ln)	Country i, year t in billions of importer currency	b
Importer price index (ln)	ln (nominal GDP) − ln (real GDP) in local currency	b
Annual exchange rate	Simple average of ln monthly rate, where latter is defined as ln ((escudo/importer currency)/200.482). The fixed conversion factor from escudo to euro is 200.482 and plays no role in the regressions	b
Firm and policy data in estimates of Tables 2–8:		
Change in number of firms(ln)	ln (number firms exporting to i in V, 1987) − ln (number firms exporting to i in V, 1985) where i is an EC-11 country and V corresponds to a NIMEXE 2-digit industry	a
Change in number of firm-varieties(ln)	ln (number varieties exported to i in V, 1987) − ln (number varieties exported to i in V, 1985), "varieties" defined as distinct 8-digit NIMEXE products exported by each firm	a
Change in exports(ln)	ln (export value to i in V, 1987) − ln (export value to i in V, 1985)	a
Pre-tariff (GATT)	ln τ where τ is 1 + ad valorem rate at product level that GATT members faced in Spain or EC-10, which is then averaged to NIMEXE 2-digit industry	c,d
Pre-tariff (Portugal)	ln τ where τ is 1 + ad valorem rate at product level Portugal faced in Spain or EC-10, which is then averaged to NIMEXE 2-digit industry	c,d
Post-tariff (Portugal)	ln τ for immediate post-agreement period that Portugal faced in Spain or EC-10, constructed as described in previous section	c,d
Applied tariff standard deviation change	Δstd (ln τ) where the standard deviation is over tariffs Portugal faced in each NIMEXE 2-digit industry; the change is between the pre- and post-tariff	f
Uncertainty	$1 - (\tau_{tiV}/\tau_{hiV})^{\sigma}$ Proportional reduction in per period profits if the tariff faced by an exporter reverts from the preferential tariff received prior to accession (pre-tariff above) to the tariff received by all nonpreferential partners (i.e., the GATT member tariff)	g
NTM share change	Change in coverage ratio measured by fraction of products in 2-digit industry subject to a specific tariff or other NTB	c,d
Specific tariff share change	Difference in the share of lines in 2-digit industry with specific tariffs between post and pre-agreement period	f

(*Continued*)

Data Appendix Table, part 1

DATA APPENDIX TABLE (*Continued*)

Variable	Description	Source
Price index proxy change	Difference in Spain's external tariff, ln(1 + ad valorem rate), between post- and pre-agreement period	f
Other data (figures and text):		
Import and export to GDP ratios	Referenced in text	h
Trade shares	Figure 1, IMF direction of trade statistics	
Export price index (ln)	1985 base chain price index for exports	h
Employment	Number of employees	i
Firm identifier (*NPC*)	Unique code to match firms between 1981–1985. Portuguese customs changed this code in 1986 and it is consistent for 1986 onwards but not between 1985 and 1986	INE
New exporter in year t	Firm exporting to a market at t but not in $t-1$ for $t < 1986$ and $t > 1986$, used in online Appendix	i
Gross entry rate in year t	(Total number new exporters in t)/(Number exporters $t-1$).	i
Gross exit rate in year t	(Number exporters with positive exports in $t-1$ and none in t)/(Number exporters with positive exports in t)	i

Notes: Additional details are in the supplementary Appendix.
[a] Authors' calculations based on INE data.
[b] IMF International Financial Statistics (IFS), volatility uses monthly data.
[c] Spain: International Customs Journal, Spain, No. 24, 16th Edition, 1984.
[d] EC: Official Journal, L342, 31.12.1979, p. 1–382.
[e] Articles of Accession, Official Journal L 302, 15/11/1985.
[f] Calculations based on tariff schedules.
[g] Authors' calculations using equation (12) with $\sigma = 3$ in the baseline regressions.
[h] Authors' calculations from Pinheiro (1997). Indices are yearly price deflators of export goods to all destinations.
[i] Authors' calculations using trade data matched to firm employment data (Quadros Pessoal) by INE.

Data Appendix Table, part 2

7 Short summary

- The paper studies whether trade-policy uncertainty deters firm export-entry investment.
- The model combines sunk export costs, heterogeneous firms, and stochastic trade policy.
- The key prediction is that uncertainty lowers entry whenever current tariffs are below the worst-case tariff.
- Portugal's 1986 EC accession is used as a credibility shock that removed uncertainty about the

future reversal of preferential market access.
- Firm entry responds positively to reductions in both applied tariffs and policy uncertainty.
- The estimated pre-accession probability of preference reversal is about 0.39 in the baseline calculation.
- Counterfactuals imply that uncertainty removal accounts for most predicted entry growth and a large share of predicted export growth.

8 Conclusion

8.1 Core conclusion

The paper's central conclusion is that credible trade policy can stimulate trade and export investment even when applied tariffs are already low. Firms care about the distribution of future policy, not only the current tariff. If exporters fear that preferential access may be withdrawn, they delay or avoid sunk entry investments. By making preferences permanent, Portugal's EC accession lowered trade-policy uncertainty and triggered export entry and export growth.

8.2 Economic intuition

The intuition is an option-value mechanism. A potential exporter must pay a sunk cost before entering. If the current tariff is low but could rise later, entering today exposes the firm to a future profit loss. Waiting preserves the option to enter after policy uncertainty is resolved. A credible agreement lowers the value of waiting and raises the entry cutoff, allowing more firms to export.

8.3 Incremental contribution revisited

Relative to standard trade-agreement studies, the paper shows that the effect of PTAs should not be evaluated only through contemporaneous tariff cuts. A PTA can also transform temporary preferences into secure market access. This credibility channel is measured directly through tariff schedules and then connected to firm-level entry.

8.4 Implications

The results help explain why unilateral trade preferences for developing countries may have weak investment effects: firms may discount them if renewal is uncertain. They also explain why PTAs sometimes generate large trade responses even when observed applied tariffs barely change. The missing variable may be a decline in trade-policy uncertainty.

8.5 Limitations

The empirical design is strongest for the Portugal accession episode, but the inferred reversal probability relies on model structure and elasticity assumptions. The analysis focuses on export entry and aggregate export values; other margins, such as pricing, product upgrading, and technology investment, are discussed but not fully modeled in the baseline empirical design.

8.6 Takeaway

A trade agreement is valuable not only because it lowers trade barriers, but also because it makes those barriers predictable. For firms facing sunk costs, credibility itself is a form of market access.